

# AI-Assisted Mineral Identification

Capstone Pre-Proposal | University of Puerto Rico at Mayagüez

Advisor: Professor Kumar | Draft for discussion

## Problem and Goal

Identifying a mineral from a photograph can be difficult because similar minerals may look alike, and lighting or specimen condition can affect their appearance. Our project will explore whether an AI-based application can support preliminary identification by suggesting likely mineral types. We aim to build a useful prototype and evaluate it on a validation set of images it has not seen during training.

## Proposed Solution

We propose a web-based application that allows users to upload a photograph with one prominent mineral specimen. The application will suggest the most likely mineral types, display reference information about those candidates, and flag predictions with low confidence. These results will be presented as suggestions rather than confirmed identifications. The first version will support a limited set of mineral classes, selected after we review the available images and discuss priorities with Professor Kumar. Recognizing several different minerals within the same photograph will be outside the initial scope.

## Technical Approach

We plan to use transfer learning, which means adapting a model that has already learned general image pattern features instead of training a new model from the beginning. We will compare two existing image-classification models, EfficientNet-B0 and ResNet-18, and select one based on its results and response time. [1]

The recommended mineral-photo dataset will be our starting point. Before training, we will check its usage permissions, image labels, duplicates and number of examples in each class. [2] A separate mineral-properties dataset can provide reference details after a prediction; these are not measurements of the photographed specimen. [3]

## High-Level Architecture

The application will have three main parts: a web-based interface for uploading photos and viewing results; a cloud-hosted prediction service that prepares each image, runs the saved model, and calculates confidence scores; and a reference data store containing information about the supported minerals. The web interface will send each uploaded image to the remote prediction service and display the results returned by it. Model training and evaluation will happen separately before the selected model is deployed to the cloud-hosted service.

## Scope and Why Heatmaps Are Excluded

We are excluding heatmaps and drilling recommendations so that image classification receives the main attention, consistent with our understanding of Professor Kumar's priorities. Mapping chemical measurements and interpreting them as exploration targets would introduce geospatial analysis, geochemistry, mineral exploration and a separate validation process. Our three-person team has limited experience in these fields and is already learning the machine-learning concepts needed for classification. Pursuing both directions would spread our effort across too many unfamiliar areas and reduce the time available to build and test either feature carefully. Heatmaps and drilling guidance are therefore left for future work as stretch goals, with appropriate domain support.

## Evaluation and Expected Deliverables

We will measure correct identifications overall and for each mineral, common errors, and response time. Photos of the same specimen will stay in one group, so final testing uses specimens not used to train or tune the model. Where possible, we will include expert-verified smartphone photos. We will also test minerals outside the supported set and other objects; low-confidence warnings cannot guarantee detection of every error.

Deliverables will include a working web-based application prototype, a comparison of the two models, and documentation of data preparation, testing and limitations. The minimum viable product (MVP) will be the smallest version meeting the class coverage and performance targets agreed with Professor Kumar.

## Open Questions for Professor Kumar

1. Which minerals are essential for the MVP? Should we cover all 15 categories listed in the recommended photo dataset, or begin with a smaller set supported by enough reliable images?
2. What minimum accuracy should the MVP reach? Should success mean that the first suggestion is correct, or that the correct mineral appears among the top three? Should there also be a minimum for each mineral, and should the target apply to new smartphone photos?
3. Can you help us obtain expert-verified test specimens or images? These will help us agree on a realistic target and check whether the application meets it.
4. Should this be a web-based application or a desktop one?

## Initial References

[1] PyTorch. [Transfer Learning for Computer Vision Tutorial](#)

[2] Florian Geillon. Kaggle. [Mineral photos](#)

[3] vinven7. Kaggle. [Comprehensive database of Minerals](#)